Module 03: Perception in Crochet Circles

How Yarn Speaks Through Touch

Learning Objectives

1. Explain how fingers read yarn weight, texture, and irregularities as sensory evidence.
2. Describe the detection of knots or inconsistencies as surprise (prediction error) in the Active Inference framework.
3. Connect hand feel and color perception in variegated yarns to interoceptive and exteroceptive sensing.

Introduction

Yarn speaks, but not in words. It speaks through the language of touch — through drag and slip, warmth and coolness, spring and drape, smoothness and texture. Every centimeter of yarn that passes through a crocheter's fingers carries information, and the crocheter's perceptual system is constantly reading, interpreting, and responding to that information.

In Active Inference, **perception** is not passive reception of data. It is an active process of inference — the brain (and the body) are constantly generating predictions about what sensory signals should arrive, and then comparing those predictions against what actually arrives. The difference is **prediction error**, and it is prediction error that drives learning, adaptation, and attention.

This module explores how yarn communicates with the crocheter through touch, sight, and even sound — and how the crocheter's perceptual system makes sense of it all.

Key Concepts

1. Reading Yarn Weight Through Touch

When a crocheter handles yarn, their fingers are performing a sophisticated measurement process. Without a scale or a label, experienced hands can distinguish between lace-weight and fingering-weight yarn, between DK and worsted, between bulky and super bulky. How?

The fingers are reading multiple sensory channels simultaneously: - **Diameter**: The thickness of the strand as it passes between thumb and forefinger - **Density**: How heavy the yarn feels for its thickness — a tightly spun worsted feels denser than a loosely spun one of the same diameter - **Resistance**: How much the yarn resists being pulled — heavier weights create more drag through the tensioning fingers - **Drape**: How the yarn hangs when a length is suspended — a heavy yarn falls straight, a light one may float or drift

Each of these sensory channels provides evidence that the crocheter's perceptual system integrates to form a belief about the yarn weight. This is Bayesian perception in action — combining multiple streams of uncertain evidence to arrive at a best estimate.

2. Surprise: Detecting Knots and Irregularities

Imagine you are crocheting steadily, working through a rhythm of single crochet stitches. Your fingers are feeding yarn at a consistent rate, and the sensory signal is smooth and predictable. Then — a bump. A knot. A sudden change in thickness.

That bump is **surprise** in the Active Inference sense. Your perceptual system was predicting a smooth, uniform signal, and what it received was a sudden deviation from that prediction. The prediction error is large, and it immediately captures your attention.

This is why knots are so noticeable even though they are tiny — they represent a large prediction error relative to the uniform signal the hands were expecting. A knot in a textured yarn (where the baseline prediction already includes some irregularity) produces less surprise than a knot in a smooth yarn, because the perceptual model already expects variation.

Other forms of surprise in yarn handling include: - **Thin spots**: Where the yarn unexpectedly narrows, creating a sudden drop in the expected diameter signal - **Joins**: Where two ends of yarn are spliced together, creating a brief change in texture or thickness - **Color changes in solid yarn**: An unexpected shift in shade that the eyes detect against the expectation of uniformity - **Moisture patches**: A spot that feels different because it absorbed moisture from your hands differently

3. Hand Feel as Interoception

Not all perception in crochet is about the yarn. Some of it is about the crocheter's own body — and this is where **interoception** comes in.

Interoception is the perception of your own internal states: fatigue, tension, comfort, pain. When you crochet, you are constantly receiving interoceptive signals: - **Hand fatigue**: The gradual buildup of tension in the muscles of the hand, wrist, and forearm - **Grip pressure**: The proprioceptive awareness of how tightly you are holding the hook and the yarn - **Posture feedback**: The sense of how your shoulders, neck, and back are positioned during extended work - **Emotional tone**: The felt sense of satisfaction, frustration, boredom, or flow that accompanies the work

These interoceptive signals are just as important as the exteroceptive signals from the yarn, because they influence how you work. As your hands fatigue, your tension changes. As your emotional state shifts, your stitch quality may vary. A skilled crocheter has learned to monitor these internal signals and adjust accordingly — taking breaks before fatigue compromises their work, stretching to maintain posture, noticing when frustration is affecting their stitches.

4. Color Perception in Variegated Yarns

Variegated yarn adds a visual dimension to the perceptual challenge. When working with a yarn that changes color along its length, the crocheter must perceive and predict not just tactile properties but also visual patterns.

Working with variegated yarn involves a unique form of perceptual prediction: - **Color sequence**: The crocheter begins to predict when the next color change will come. After several rows, they develop expectations about the repeat length. - **Pooling patterns**: The way colors distribute across the fabric depends on the stitch count per row. A crocheter who understands this can predict — and manipulate — whether colors will pool, stripe, or argyle. - **Contrast**

detection: The eye constantly compares adjacent colors, detecting where transitions are smooth and where they are jarring. High-contrast transitions generate more visual surprise.

This is multimodal perception — the integration of tactile and visual information to form a unified understanding of the material being worked. The hands feel the yarn while the eyes watch the fabric grow, and both streams of evidence are combined into a single, coherent perceptual model.

5. Active Sensing: Seeking Information

Perception in Active Inference is not just about passively receiving signals — it is about **actively seeking** information to reduce uncertainty. Crafters do this constantly:

- **The squeeze test**: Rolling yarn between the fingers to actively sample its texture and compression
- **The snap test**: Pulling a length of yarn taut and releasing to hear and feel its elasticity
- **The drape test**: Holding a length of yarn up and letting it hang to observe its behavior under gravity
- **The burn test**: Touching a flame to a fiber end to identify the fiber type by smell and behavior (protein fibers smell like burning hair, plant fibers like burning paper, synthetics melt)

Each of these is an **epistemic action** — an action taken not to change the world but to gain information about it. In Active Inference terms, these are actions that reduce expected free energy by resolving uncertainty about the yarn's hidden states.

The Free Energy Principle and Perception

In the Free Energy Principle framework, perception is the process of minimizing prediction error by updating the internal model. When the yarn feels as expected, prediction error is low and perception is smooth and effortless. When something unexpected happens — a knot, a color change, a new fiber type — prediction error increases, and the perceptual system must work harder.

This is why familiar materials feel "right" — there is a deep perceptual comfort in working with a yarn whose properties match your predictions. And it is why new materials can feel unsettling or exciting — the prediction errors they generate demand attention and model updating.

The crocheter's perceptual skill grows with experience. Each new yarn handled, each knot encountered, each color transition observed adds to the generative model. Perception becomes richer, more nuanced, and more confident — not because the senses change, but because the predictions they are compared against become more refined.

Conclusion

Yarn speaks through touch, sight, weight, and drag. The crocheter's perceptual system is constantly reading these signals, comparing them against predictions, and generating the rich sensory experience of working with fiber. Knots are surprise, smooth yarn is confirmed expectation, and the skilled crocheter's perceptual world is a constant flow of evidence and inference. In the next module, we will explore how this perceptual information feeds into cognition — the thinking that happens in the hands.

Connections

- **Previous Module**: Agents — Hands That Know the Yarn
- **Next Module**: Cognition — When Your Hands Think Ahead
- **Course Home**: Fiber & Flow